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MS2130

USB 3.0 High-Definition Video Capture

Chip Datasheet

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1. Basic Introduction

The MS2130 is a USB 3.0 high-definition video and audio capture chip that integrates a USB 3.0 device controller and data transfer function. The MS2130 includes a transceiver module and an audio/video processing module. It can transmit HD input audio and video signals to a PC via a USB 3.0 interface. Preview or capture on smartphones and tablets. The MS2130 output supports both YUV422 and MJPEG modes and is compatible with Windows. Android and macOS systems.

2. Functional characteristics

HD RX

Supports RGB444, YCBCR422, YCBCR444, YCBCR420 color space

Supports deep color 24/30/36-bit modes.

Supports CEA-861-E/CEA-861-F video formats.

Maximum input resolution: 4K@30Hz

audio

ÿ USB audio output is 48kHz stereo by default.

USB 3.0

Built-in USB 3.0 Device

ÿ Compatible with USB 2.0 Device mode

ÿ USB video complies with UVC1.0 protocol

USB audio complies with UAC1.0 protocol.

ÿ USB video output supports both YUV422 and MJPEG.

Types of modes

ÿ USB video output resolution

Default maximum resolution 1920×1080@60Hz

Operating system support

Windows XP/7/8/10/11

ÿ Android 5.0 or above

ÿ Linux

macOS

Clock processing

External 24MHz passive crystal oscillator

Built-in PLL circuit

Reset

Built-in power-on reset circuit

Packaging

ÿ QFN-64 plastic seal (9mm×9mm)

ÿ Compliant with RoHS standards

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3. Application Scenarios

High-definition video capture card

ÿ Security monitoring

ÿ Medical imaging

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5. Functional Block Diagram

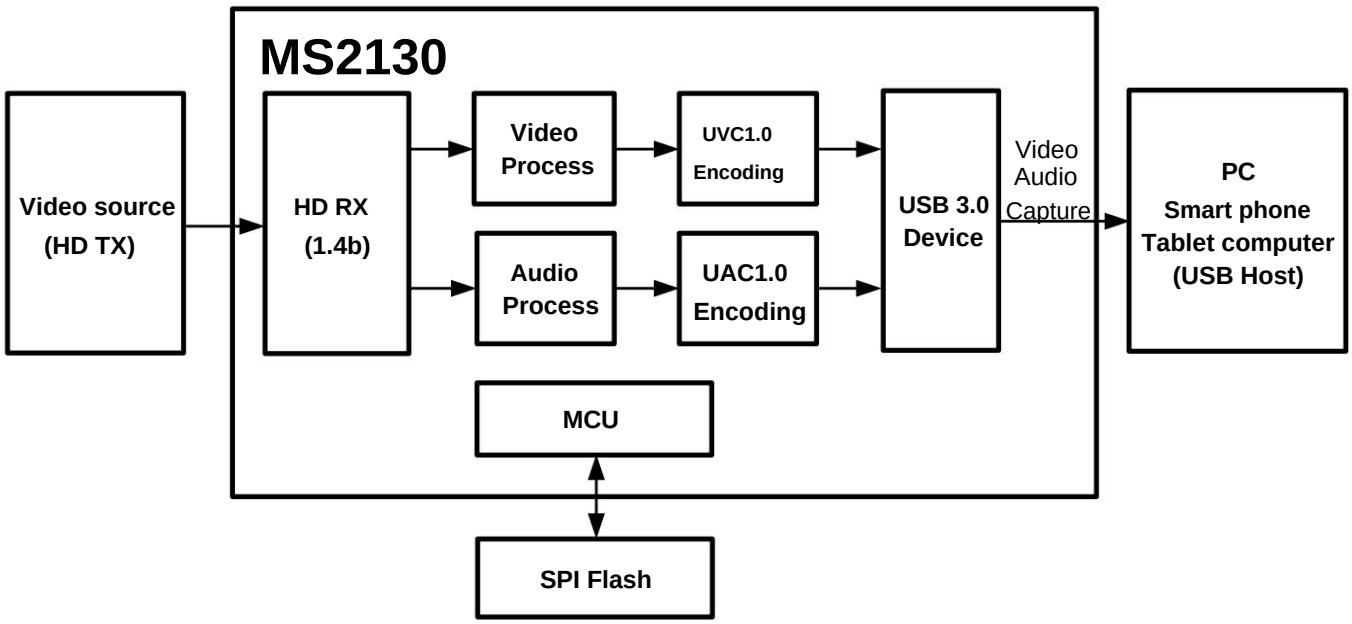


Figure 1. Functional Block Diagram

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6. Function Description

6.1 GPIO

The MS2130 has 6 programmable GPIOs, whose functions can be customized via software.

6.2 HD RX EDID Description

The MS2130 supports HD RX, HPD processing, and custom EDID.

Table **6.1** describes the default **EDID** for HD RX as follows:

category	describe
Display Product Name: Best	HD to USB
Resolution 1	1080P@60Hz, 148.5MHz
List of supported video resolutions	640×480@60Hz 800×600@60Hz 1024×768@60Hz 1152×864@60Hz 1280×720@60Hz 1280×800@60Hz 1280×960@60Hz 1280×1024@60Hz 1440×900@60Hz 1600×900@60Hz 1680×1050@60Hz 1920×1080@60Hz 3840×2160@30Hz 3840×2160@50Hz 3840×2160@60Hz 4096×2160@30Hz 4096×2160@50Hz 4096×2160@60Hz
Video color space support	RGB, YUV444, YUV422, YUV420
Deep color support	30 bits, 36 bits
Audio format support	Format: L-PCM; Sampling rate: 32kHz, 44.1kHz, 48kHz; Number of channels: Maximum two channels; Bit depth: 16bit, 20bit, 24bit

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6.3 USB video output

The MS2130 supports video capture in USB 3.0 & USB 2.0 modes, conforms to the UVC 1.0 protocol, and supports YUV422 and MJPEG.

Two video output formats are supported, with a default maximum output resolution of 1920×1080@60Hz.

The MS2130 USB default output resolution list is as follows. Users can customize the output resolution by modifying the software configuration.

(Supports up to 4K@15Hz).

Table **6.2** Default Output Resolution

USB 3.0 mode:	
YUV422:	MJPEG:
1920×1080@60Hz/50Hz/30Hz/20Hz/10Hz 1600×1200@60Hz/50Hz/30Hz/20Hz/10Hz 1360×768@60Hz/50Hz/30Hz/20Hz/10Hz 1280×1024@60Hz/50Hz/30Hz/20Hz/10Hz 1280×960@/60Hz/50Hz/30Hz/20Hz/10Hz 1280×720@60Hz/50Hz/30Hz/20Hz/10Hz 1024×768@60Hz/50Hz/30Hz/20Hz/10Hz 800×600@60Hz/50Hz/30Hz/20Hz/10Hz 720×576@60Hz/50Hz/30Hz/20Hz/10Hz 720×480@60Hz/50Hz/30Hz/20Hz/10Hz 640×480@60Hz/50Hz/30Hz/20Hz/10Hz	1920×1080@60Hz/50Hz/30Hz/20Hz/10Hz 1600×1200@60Hz/50Hz/30Hz/20Hz/10Hz 1360×768@60Hz/50Hz/30Hz/20Hz/10Hz 1280×1024@60Hz/50Hz/30Hz/20Hz/10Hz 1280×960@/60Hz/50Hz/30Hz/20Hz/10Hz 1280×720@60Hz/50Hz/30Hz/20Hz/10Hz 1024×768@60Hz/50Hz/30Hz/20Hz/10Hz 800×600@60Hz/50Hz/30Hz/20Hz/10Hz 720×576@60Hz/50Hz/30Hz/20Hz/10Hz 720×480@60Hz/50Hz/30Hz/20Hz/10Hz 640×480@60Hz/50Hz/30Hz/20Hz/10Hz
USB 2.0 mode:	
YUV422:	MJPEG:
1920×1080@10Hz/5Hz 1600×1200@10Hz/5Hz 1360×768@15Hz/8Hz 1280×1024@15Hz/8Hz 1280×960@15Hz/8Hz 1280×720@20Hz/10Hz 1024×768@20Hz/10Hz 800×600@30Hz/20Hz/10Hz 720×576@50Hz/25Hz/20Hz/10Hz 720×480@60Hz/30Hz/20Hz/10Hz 640×480@60Hz/30Hz/20Hz/10Hz	1920×1080@50Hz/30Hz/25Hz/20Hz/10Hz 1600×1200@60Hz/50Hz/30Hz/20Hz/10Hz 1360×768@60Hz/50Hz/30Hz/20Hz/10Hz 1280×1024@60Hz/50Hz/30Hz/20Hz/10Hz 1280×960@/60Hz/50Hz/30Hz/20Hz/10Hz 1280×720@60Hz/50Hz/30Hz/20Hz/10Hz 1024×768@60Hz/50Hz/30Hz/20Hz/10Hz 800×600@60Hz/50Hz/30Hz/20Hz/10Hz 720×576@60Hz/50Hz/30Hz/20Hz/10Hz 720×480@60Hz/50Hz/30Hz/20Hz/10Hz 640×480@60Hz/50Hz/30Hz/20Hz/10Hz

6.4 USB audio output

The MS2130 USB audio output conforms to the UAC1.0 protocol and defaults to 48kHz stereo output, which can be configured via software.

Resolution audio output.

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7. Pin Diagram

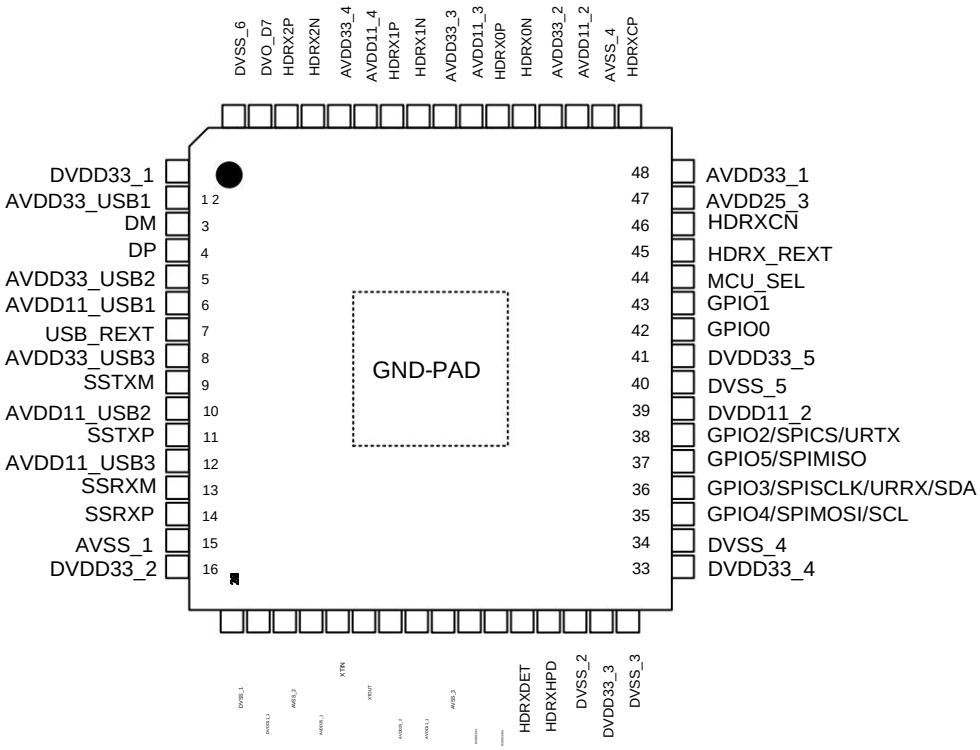


Figure 2. Pin Diagram

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8. Pin Description

Table **8.1** Pin Description

Pin Name	Pin #	type	describe
General **I/O**			
GPIO0	42	General purpose digita	I/O
GPIO1	43	General purpose digita	I/O
GPIO2/SPICS/URTX	38	Digital Input/Output	General purpose digital I/O, multiplexing SPI Master interface CS PIN and UART TX PIN; If an external SPI interface Flash is connected, the chip will power on after... Automatically configured as SPICS function
GPIO3/SPISCLK /URRX/SDA	36	Digital Input/Output	General purpose digital I/O, multiplexing SPI Master interface SCLK PIN, UART RX PIN, and I2C Master SDA PIN; If an external SPI interface Flash is connected, the chip will power on after... It will be automatically configured to use the SPISCLK function.
GPIO4/SPIMOSI/SCL	35	Digital Input/Output	General purpose digital I/O, multiplexing SPI Master interface MOSI PIN and I2C Master SCL PIN; If an external SPI interface Flash is connected, the chip will power on after... It will be automatically configured as a SPIMMOSI function.
GPIO5/SPIMISO	37	Digital Input/Output	General purpose digital I/O, multiplexing SPI Master interface MISO PIN If an external SPI interface Flash is connected, the chip will power on after... Automatically configured for SPIMISO functionality
USB			
DP	4	Digital input/output USB 2.0 differential positive terminal signal	
DM	3	Digital input/output USB 2.0 differential negative terminal signal	
USB_REXT	7	The analog output is connected to ground via a 200-ohm resistor.	
SSTXP	11	Digital output USB 3.0 differential output positive terminal signal	
SSTXM	9	Digital output USB 3.0 differential output negative terminal signal	
SSRXP	14	Digital input USB 3.0 differential input positive signal	
SSRXM	13	Digital input USB 3.0 differential input negative terminal signal	
HD RX			
HDRXCN	46	Digital input HD receiver differential clock input negative signal	
HDRXCP	49	Digital input HD receiver differential clock input positive signal	
HDRX0N	53	Digital input HD receiver differential channel 0 data input negative signal	
HDRX0P	54	Digital input HD receiver differential channel 0 data input positive signal	
HDRX1N	57	Digital input HD receiver differential channel 1 data input negative signal	
HDRX1P	58	Digital input HD receiver differential channel 1 data input positive signal	
HDRX2N	61	Digital input HD receiver differential channel 2 data input negative signal	
HDRX2P	62	Digital input HD receiver differential channel 2 data input positive signal	
HDRX_REXT	45	Connect a 1.6kÿ resistor to ground for the analog output.	

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Pin Name	Pin #	Type	describe
HDRXHPD	29	Number Output (Tolerates 5V)	HD receiver hot-plug signal output
HDRXDET	28	Numeric input (Tolerates 5V)	HD receiver 5V input detection
RXDDCSDA	27	Digital Input/Output (Tolerates 5V)	HD receiver displays data channel serial data bus
RXDDCSCL	26	Numeric input (Tolerates 5V)	HD receiver displays data channel serial clock bus
Debug **PIN**			
MCU_SEL	44	Numeric input debugging PIN, default is floating.	
DVO_D7 crystal	63	Digital output debugging PIN, floating by default.	
oscillator			
XTIN	twenty one	Analog input connected to 24MHz passive crystal oscillator input	
XTOUT	twenty two	Analog output connected to 24MHz passive crystal oscillator output	
System power and ground			
AVDD33_USB1~3	2,5,8	Power USB module with 3.3V power supply	
AVDD33_1~4	48,52,56,60	Simulates 3.3V power supply	
DVDD33_1~5	1,16,31,33,41	Power digital I/O 3.3V power supply	
AVDD25_1~3	20,23,47	Simulates 2.5V power supply	
AVDD11_USB1~3	6,10,12	Power USB module is powered by 1.1V.	
DVD11_1~2	18,39	Digital 1.1V power supply	
AVD11_1~4	24,51,55,59	Simulate 1.1V power supply	
AVSS_1~4	15,19,25,50	Earth simulation	
DVSS_1~6	17,30,32,34, 40,64	Digital land	
GND-PAD		Ground handling simulation	

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9. Electrical characteristics

9.1 Limiting Parameters

Table **9.1** Limiting Electrical Parameters

	symbol	numerical values	unit
Parameter limit operating voltage (AVDD33/DVDD33/DVDDIO)	VDD33	3.63	V
Limiting operating voltage (AVDD25)	VDD25	2.75	V
Limiting operating voltage (AVDD11/DVDD11)	VDD11	1.21	V
Ambient operating temperature	TA	0 to +70	°C
Storage temperature	Tsto	-65 to +150	°C
Limiting junction temperature	Tjmax	125	°C

Note: If the operating conditions of the device exceed the above-mentioned "limiting parameters," it will cause permanent damage to the device. Only when the device operates within the specified limits...

Functionality can only be guaranteed when the device operates within the range specified in the manual. Operating the device under the listed limiting parameters will affect its performance.

reliability.

Table **9.2** Thermal resistance parameters

parameter	symbol	numerical values	unit
Junction-to-ambient thermal resistance (JEDEC 4-layer plate)	RÿJA	36	°C/W
Junction-to-board thermal resistance	RÿJB	6	°C/W
Junction-to-case thermal resistance	RÿJC	11	°C/W

9.2 Electrical Characteristics

Table **9.3** DC Parameters

parameter	Minimum value, typical value, maximum value, unit				Test conditions
VDD33	3.135	3.3	3.465	Fu	Ground = 0V, Temperature = +25°C Unless otherwise stated
VDD25	2.375	2.5	2.625	Fu	
VDD11	1.045	1.1	1.155	Fu	
Input low voltage	-0.3		0.8	Fu	

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	Minimum value, typical value, maximum value, unit				Test conditions
Parameter input:	2.0		3.6	Fu	
high voltage; output:			0.4	Fu	
low voltage; output: high voltage	2.4			Fu	
Input low voltage (5V IO tolerance)	-0.3		0.8	Fu	
Input high voltage (5V IO tolerance)	2.0		5.5	Fu	
Output low voltage (5V IO tolerance)			0.4	Fu	
Output high voltage (5V IO tolerance)	2.4			Fu	

Note: VDD33 includes AVDD33_USB, DVDD33, and AVDD33.

VDD11 includes AVDD11_USB, DVDD11, and AVDD11.

5V compliant IOs include HDRXHPD, HDRXDET, RXDDCSDA, and RXDDCSCL.

9.3 Power Consumption in Different Modes

Table **9.4** Power Consumption in Different Operating States

Test conditions: Ground = 0V, Temperature = +25°C.

parameter	**UVC ON** Input **1080P@60Hz** Output **1080P@60Hz**	**UVC OFF** **HD** not connected	**USB** sleep	unit
VDD33	0.389	0.049	0.019	watt
VDD25	0.077	0.01	0.005	watt
VDD11	0.146	0.055	0.012	watt

Table **9.5** Current in different modes

The USB input is fixed at 1920×1080@60Hz, YUV mode. Typical current values for different input modes are shown. Test conditions: Ground = 0V, Temperature = +25°C.

parameter	**4K@30Hz**	**1080P@60HZ**	**720P@60Hz**	unit
AVDD33	107	93	79	milliampere
DVDD33	28	31	twenty two	milliampere
AVDD25	31	30	19	milliampere
AVDD11	96	80	70	milliampere
DVDD11	74	66	58	milliampere

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10. Typical Application Circuits

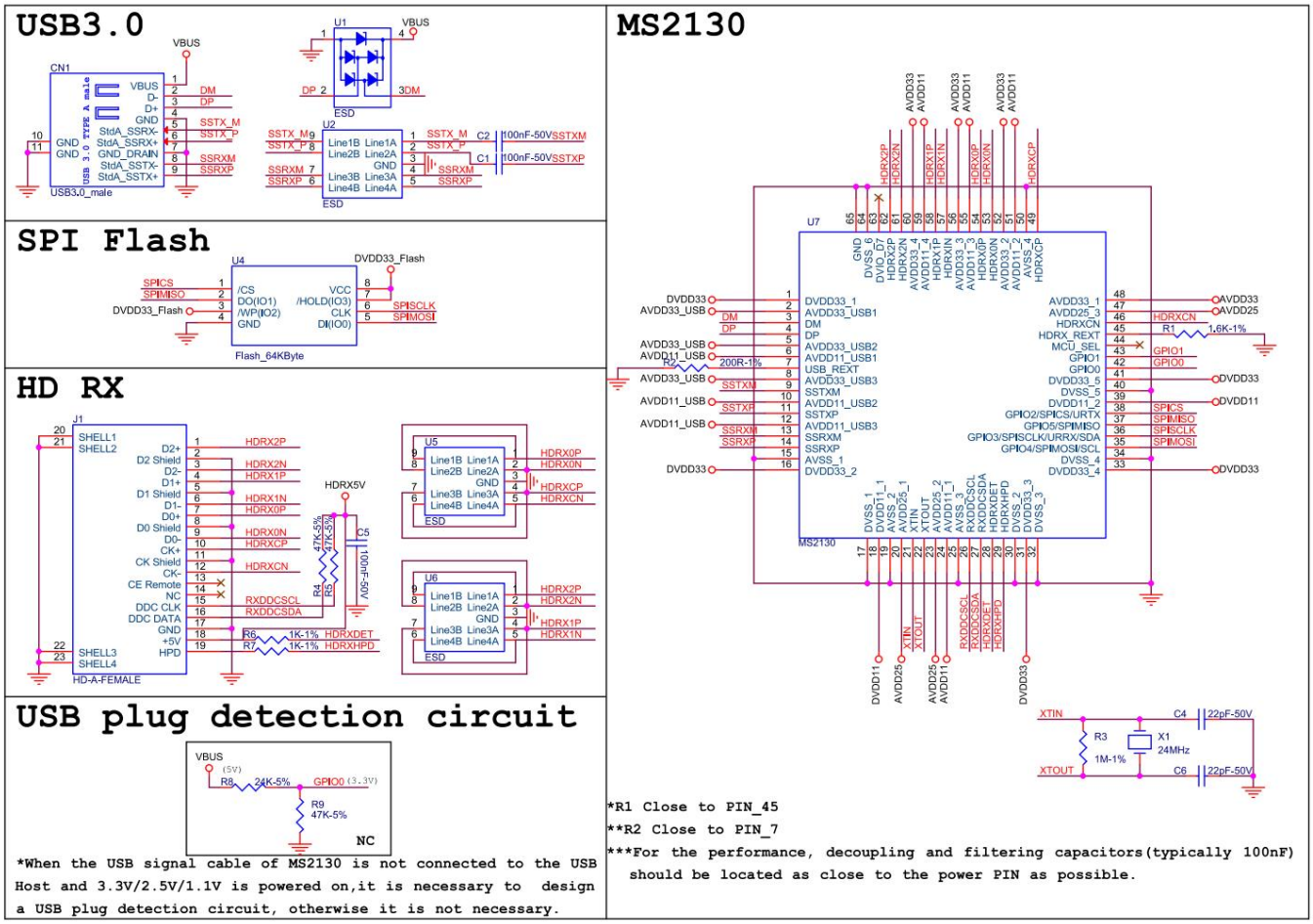


Figure 3. Typical application circuit diagram

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11. PCB Design Specifications

11.1 Power/Ground

The width of power traces must correspond to the current flowing through them; the current values for each power module can be found in the electrical characteristics section. (This is part of the PCB design process.)

Power/ground traces should be as wide as possible; the actual trace width is recommended to be no less than the reference values in the table below.

The filter capacitor for the chip's power supply needs to be close to the chip's power pins. Power must pass through the capacitor before entering the chip's pins.

The grounding point needs to be drilled nearby to connect to the power supply.

Table **11.1** Correspondence between power trace width and current value

Maximum current carrying capacity (mA)	Recommended line width (mil)
50	ÿ12
100	ÿ16
200	ÿ20
400	ÿ30
Note: All parameters above are based on a copper foil weight of 1 ounce.	

11.2 Clock

The crystal oscillator should be placed close to the chip. Wiring should be as short as possible and ground plane protection is required. Avoid routing on the layer below the crystal oscillator to ensure a complete ground plane.

11.3 USB signal cable

The impedance matching requirement for USB SSTXP/M, SSRXP/M, and DP/M differential lines is 90 ± 10% Ohm. Differential line routing should be as...

For short lengths, do not exceed one pair of vias; the inter-line tolerance should be within 5 mils; the total length is recommended not to exceed 2000 mils. Differential lines need to be grounded.

In this case, other signal lines should not be placed near the ground plane (3W principle), and no wiring should be laid on the next layer to ensure a complete ground plane.

11.4 HD signal cable

For HD differential lines, impedance matching is required to be 100 ± 10% Ohm. Differential line traces should be as short as possible, ideally not exceeding two pairs of vias.

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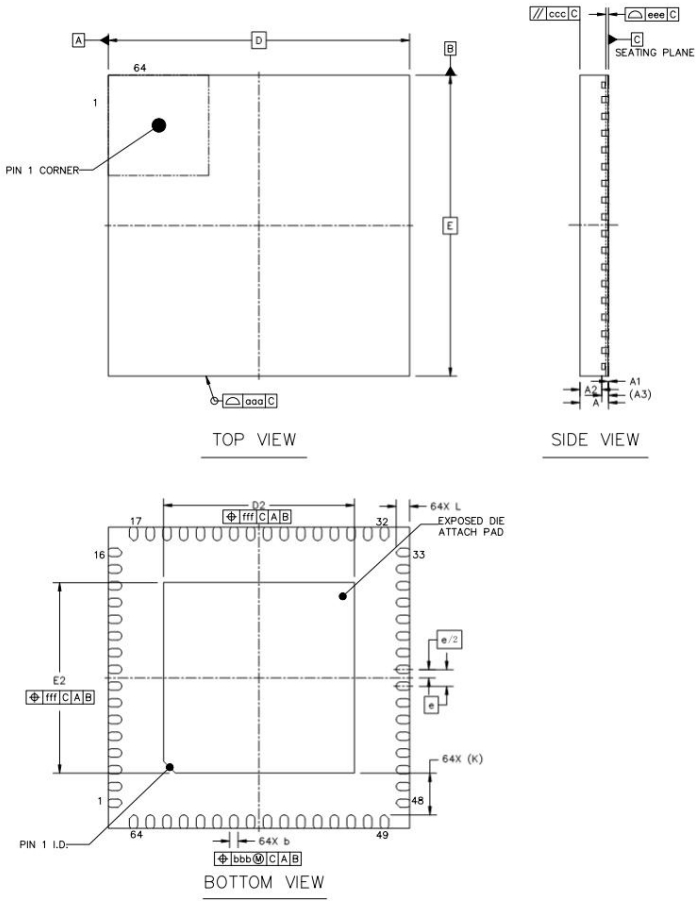
Internal tolerance within 5 mil, and group tolerance within 15 mil. Differential lines require grounding and must not be near other signal lines (3W original).

Then, do not route on the next layer to ensure a complete ground plane. ESD devices, common-mode inductors, and series resistors should be placed close to the HD interface, following the correct placement.

The order is: HD interface - ESD device - common mode inductor - resistor.

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12. Encapsulation Information



		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.8	0.85	0.9
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.65	---
L/F THICKNESS		A3		0.203 REF	
LEAD WIDTH		b	0.2	0.25	0.3
BODY SIZE	X	D		9 BSC	
	Y	E		9 BSC	
LEAD PITCH		e		0.5 BSC	
EP SIZE	X	D2	5.6	5.7	5.8
	Y	E2	5.6	5.7	5.8
LEAD LENGTH		L	0.3	0.4	0.5
LEAD TIP TO EXPOSED PAD EDGE		K		1.25 REF	
PACKAGE EDGE TOLERANCE		aaa		0.1	
MOLD FLATNESS		ccc		0.1	
COPLANARITY		eee		0.08	
LEAD OFFSET		bbb		0.1	
EXPOSED PAD OFFSET		fff		0.1	

Figure 4. QFN64 package block diagram

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13. Chip Identification

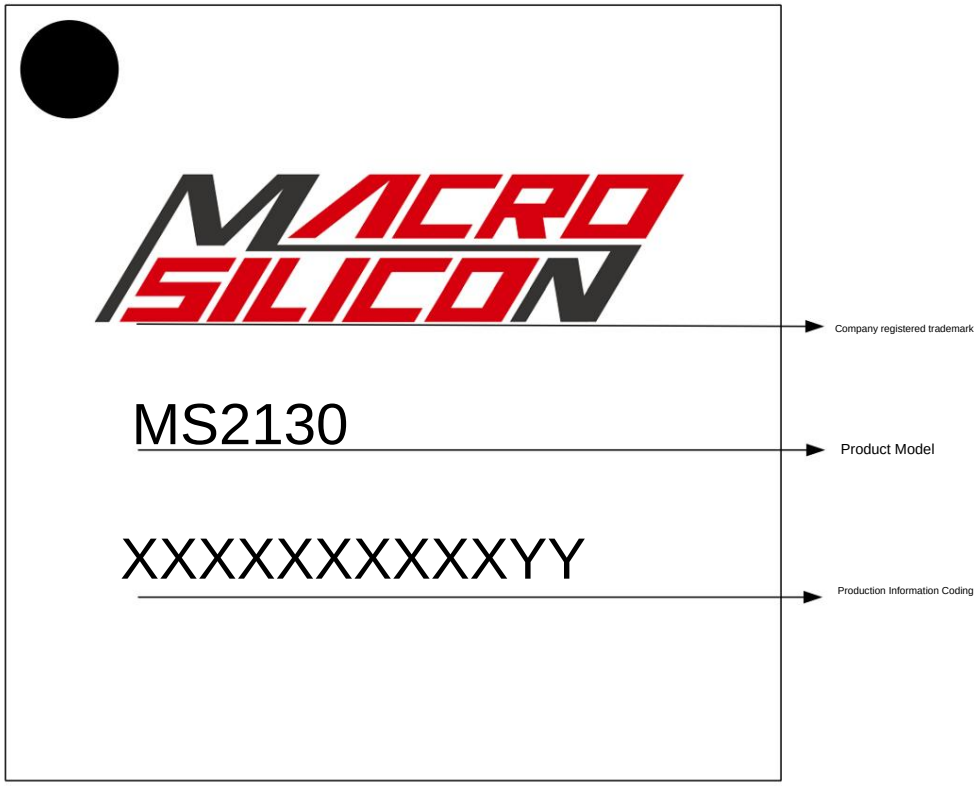


Figure 5. Chip Identification

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14. Packaging Information

14.1 Packaging Information

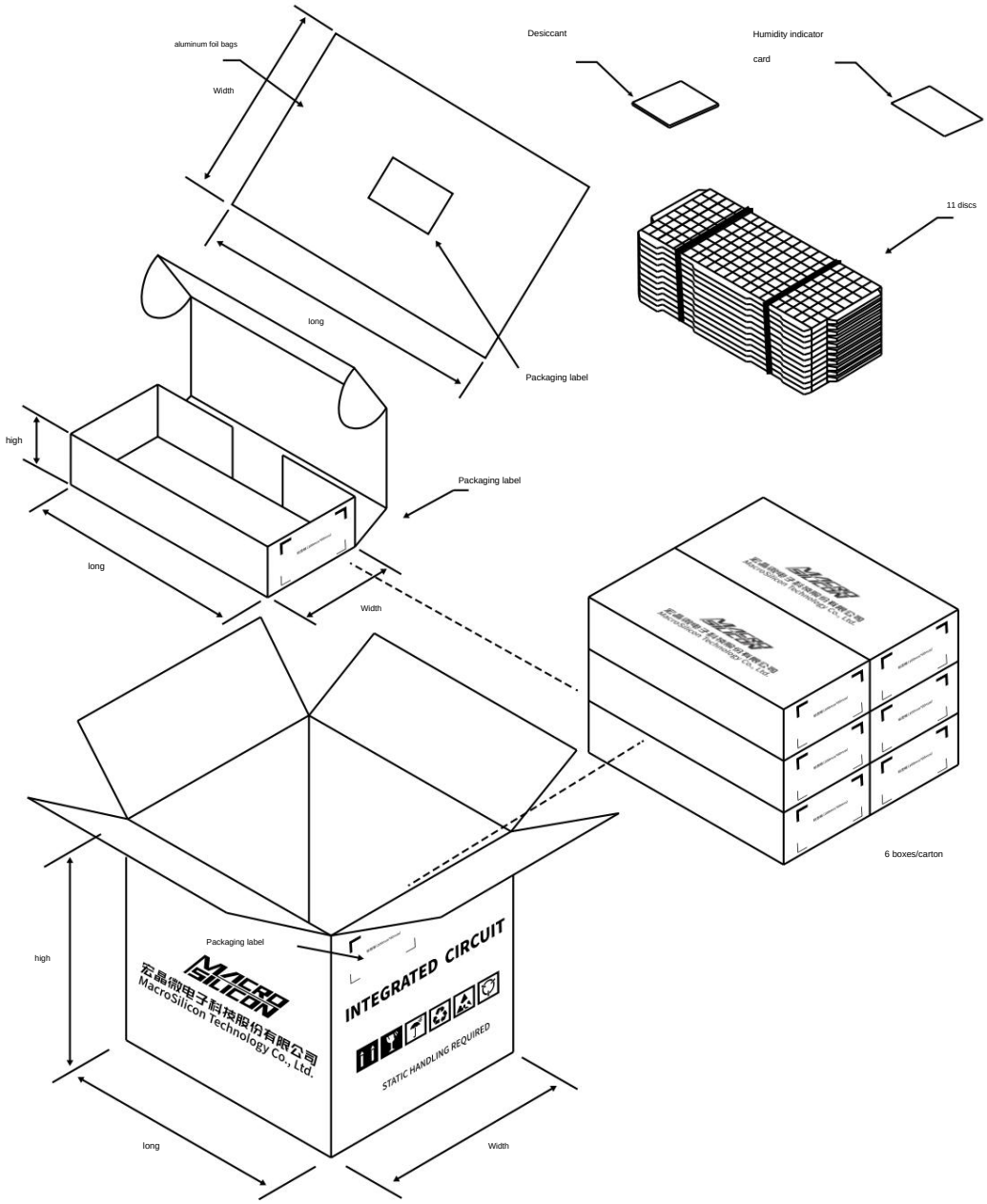


Figure 6. Packaging Infographic

Table **14.1** Packaging Carton Dimensions

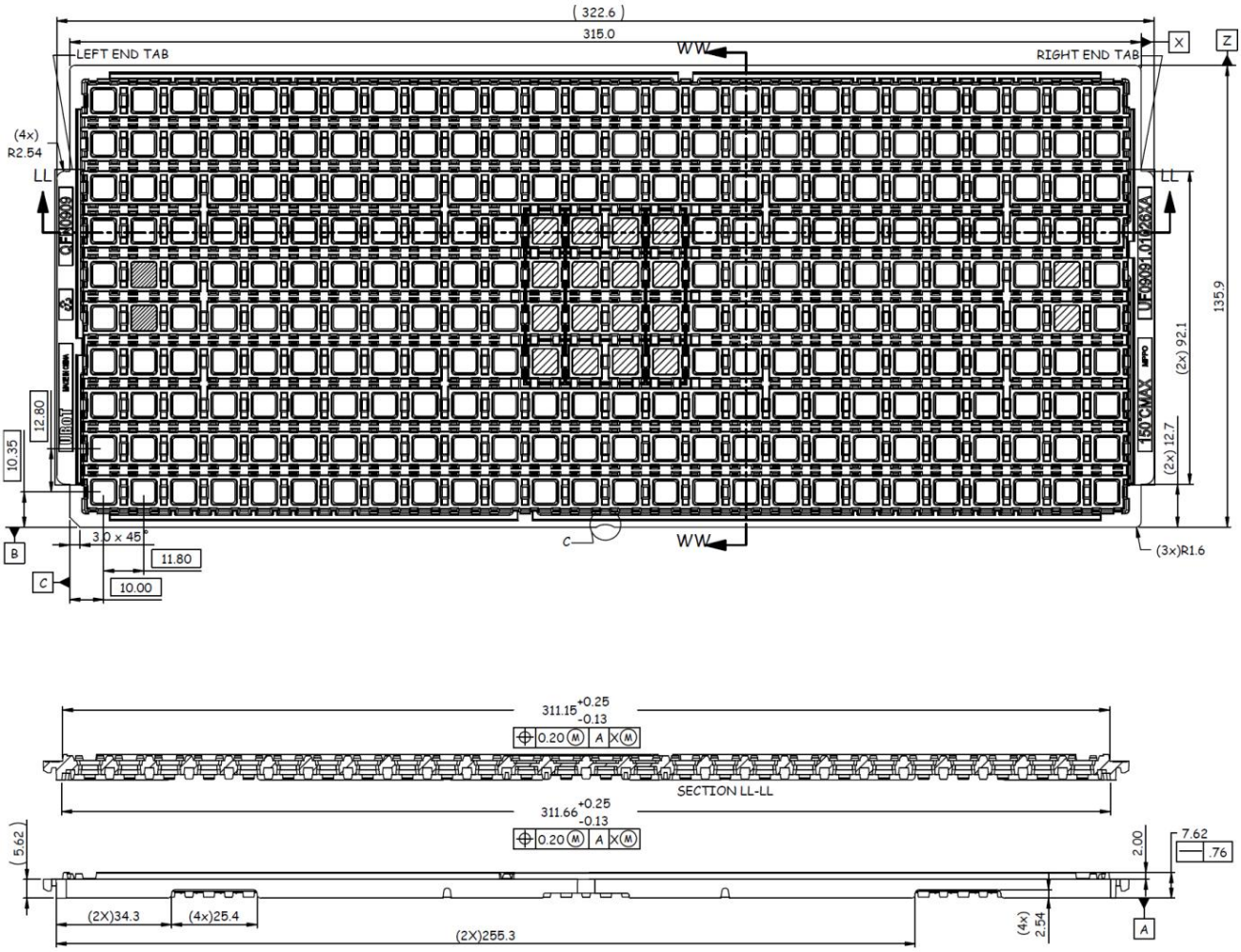
Packaging box dimensions (unit: mm)	
Inner box	370(L)*155(W)*85(H)
and outer box	390(L)*330(W)*280(H)

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Table **14.2** Packaging Standards

Package outline	Number per **TRAY** disk **(Unit: PCS)**	Quantity per inner box **(Unit: PCS)**	Quantity per outer carton **(Unit: PCS)**	Ratio of inner and outer boxes
QFN64 (9X9)	260	2600	15600	6:1

14.2 Tray size information



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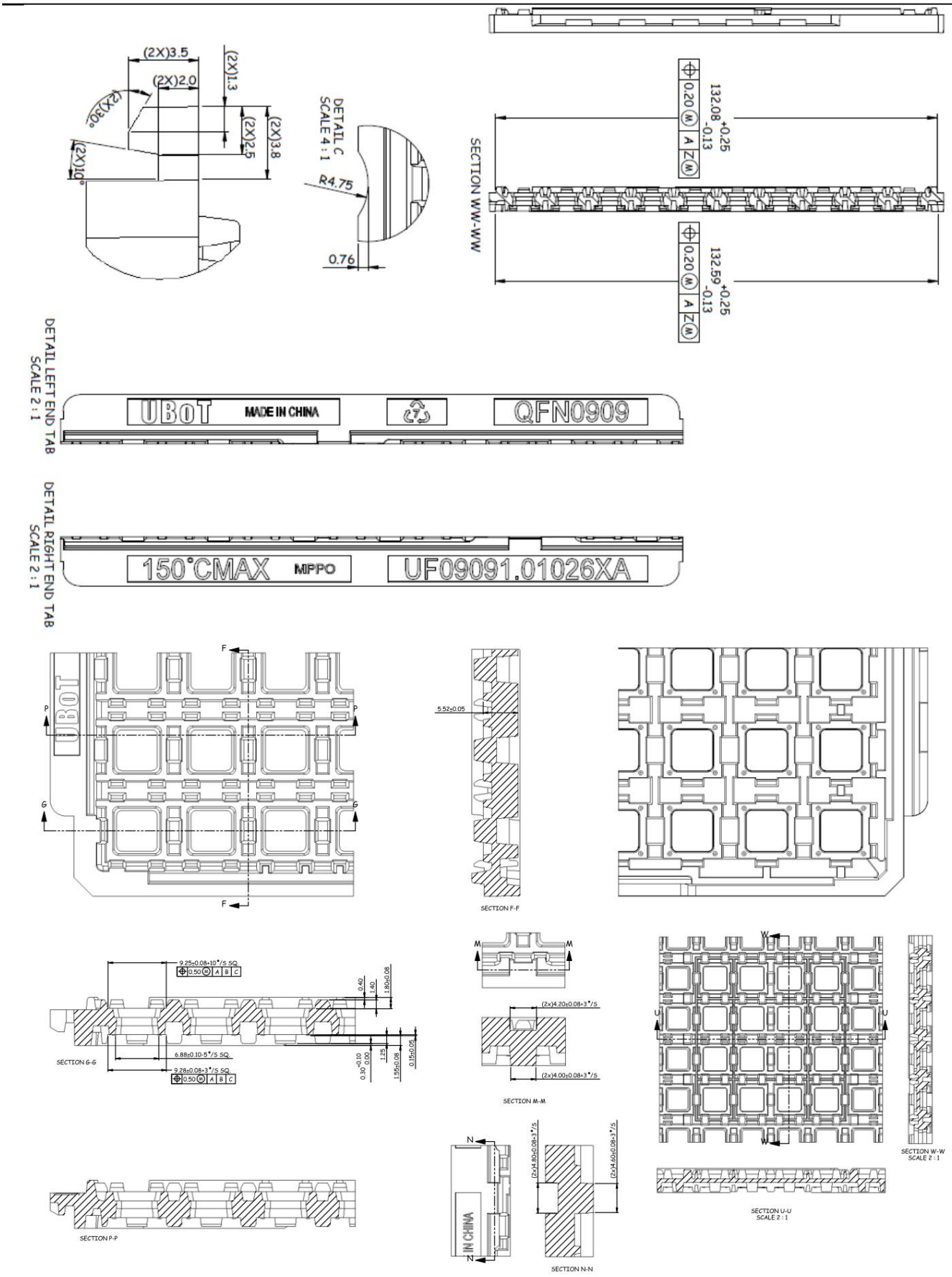


Figure 7. Tray Dimensions (Page 21 of 23) Tel: 0551-65317870

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15. Reflow soldering temperature specifications

Table **15.1** Description of Reflow Soldering Temperature Profiles

Reflow soldering	**Pb-Free Assembly** 60~120
temperature profile: Preheating time	seconds (150~200°C)
(Tsmin ~ Tsmax), Liquid	217°C
temperature (TL) , Peak	260°C (+5/-0°C)
temperature (TP), Rise rate (TL	ÿ 3.0°C/second
~ TP) , Hold time (above 217°C), Hold	for 60~150 seconds
time within 5°C of peak temperature (above 255°C), Fall rate (TP ~ TL) 30~40 seconds	
	ÿ 6.0°C/second
Time from 25°C to peak temperature	ÿ 8 minutes

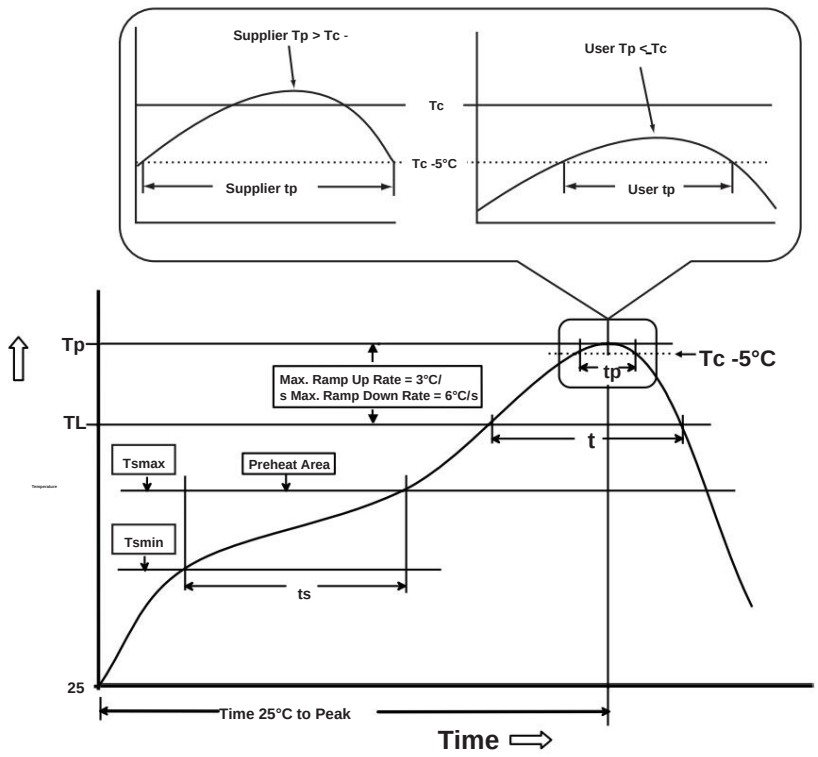


Figure 8. Reflow soldering temperature profile

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16. Version History

date	Version	Author	Remark
2021-10	V1.0	MJ Du	first edition
2023-04	V1.1	MJ Du	Modified description
2023-11	V1.2	MJ Du	Added PCB design specifications, and modified the HD section.

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